

Verification and Progress on Purity of Pure

A machine-checked report on Goodman’s project notes

27 July 2026; updated 28 July 2026

Purpose. This report records what we have verified in Jeremy Goodman’s July 2026 project notes on the consistency of Purity of Pure with Bacon’s logical combinatorialism. It has two aims. First, it separates the claims that have been verified exactly from claims that require qualifications, repairs, or further formulation. Second, it states the new theorems proved during the formalization that genuinely narrow the open consistency problem.

Bottom line. Goodman’s central consistency question remains open. Isabelle nevertheless settles the determinate subsidiary claims in the notes: each is proved, refuted and repaired, or assigned the explicit qualification under which it is true. In particular, the formalization identifies and repairs a modal gap in T3 and a false countability argument in M5; verifies the four T6 contradiction routes and the T9 cardinal dichotomy from their explicit premises; completes the rebuilt M5 model containing the displayed exotic operator; and isolates the assumptions still needed for several of the remaining model-theoretic claims. It also proves that Recombination together with zeroary Exhaustion and unique proposition-level fundamentality supplies the fun' witness used in T6. Hence T6 has the following exact semantic consequence: any model of the repaired background theory plus PP must falsify global L2 or global TU. That consequence sharply constrains a positive answer, but it does not decide whether such a model exists. We have also settled Goodman’s proposed calibration of L2 in Bacon’s appendix model: a closed logical operator Z refutes L2 when the pure unary operators are the denotations of closed terms containing only logical vocabulary. This is a substantive result about the notes, although Bacon’s model still fails PP and therefore does not answer the consistency question.

1 Formal setting and verification standard

The formal development represents Bacon’s typed language and proof system directly in Isabelle/HOL. It includes Bacon’s system H , classical and Boolean identity principles, the Rule of Equivalence, vector Equivalence, the definition of necessity by identity with truth, and the additional purity/fundamentality vocabulary used in Goodman’s notes. The development distinguishes object-language derivability from validity in a particular model. This matters throughout: a semantic argument valid in Bacon’s appendix model is not thereby a

derivation in the theory, and an Isabelle theorem with explicit premises does not show that those premises hold in the intended model.

The complete audit checks the theorem objects supporting the claims summarized below. Fresh builds pass. None of these proofs contains an admitted lemma, an untrusted proof step, or an undischarged assumption beyond the assumptions stated in the theorem itself. The type restriction carried by the general cardinal theorems in T9 is part of their formally checked statement, not a further mathematical premise.

2 Background definitions and two important distinctions

At proposition type, Goodman's definable surrogate for fundamentality is

$$\text{fun}'(p) \stackrel{\text{def}}{=} \forall X Y ((\text{Pure}(X) \ \& \ \text{Pure}(Y) \ \& \ Xp = Yp) \rightarrow X = Y).$$

Let G be the pure reversible unary operators, and let

$$X \approx Y \stackrel{\text{def}}{=} \exists Z \in G (X = Y \circ Z).$$

Weak L2 says that if X and Y are pure, p and q are fun' , and $Xp = Yq$, then $X \approx Y$.

Two distinctions uncovered by the formalization are central.

Recombination versus Exhaustion. Unary Recombination proves a pointwise modal identity principle. It does not, by itself, permit necessitating the universal closure required by the usual QSS derivation. Zeroary Exhaustion supplies exactly that missing step. With unique proposition-level fundamentality, the repaired package derives the existence of a fun' proposition.

Conditional model-theoretic results. A formally verified theorem may still have explicit premises concerning Bacon's function domains, invariance, or QSS. We therefore distinguish a verified conditional theorem from a proof that all of its premises hold in Bacon's particular appendix model.

3 Goodman's L2 principles

Goodman's July notes state one principle bearing the label L , in two forms. Weak L2 is

$$(L2) \quad \text{Pure}(X) \ \& \ \text{Pure}(Y) \ \& \ \text{fun}'(p) \ \& \ \text{fun}'(q) \ \& \ Xp = Yq \implies X \approx Y.$$

The strong form requires the same reversible operator to witness both the relation between the operators and the relation between their inputs:

(strong L2)

$$\text{Pure}(X) \ \& \ \text{Pure}(Y) \ \& \ \text{fun}'(p) \ \& \ \text{fun}'(q) \ \& \ Xp = Yq \implies \exists Z \in G (X = Y \circ Z \ \& \ q = Zp).$$

The notes contain no separate principle labeled L1. Accordingly, the plural “L principles” below refers to weak and strong L2.

3.1 Object-language verification

Both forms have been encoded as closed, well-typed object-language formulas. Isabelle verifies all four uses made in T6:

$$\begin{aligned} T_0 + \text{PP} + \exists \text{fun}' + \text{L2} + \text{Inv} &\vdash \perp, \\ T_0 + \text{PP} + \exists \text{fun}' + \text{L2} + \text{WI} &\vdash \perp, \\ T_0 + \text{PP} + \exists \text{fun}' + \text{L2} + \text{TU} &\vdash \perp, \\ T_0 + \text{PP} + \text{strong L2} + \text{RS} &\vdash \perp. \end{aligned}$$

The repaired Recombination-Exhaustion bridge derives the required $\exists \text{fun}'$ premise from Recombination, zeroary Exhaustion, and unique proposition-level fundamentality. Isabelle also verifies the uses of L2 in T7-T9, including uniqueness of kind in T8 and the conditional cardinal dichotomy in T9.

3.2 Refutation of L2 in Bacon’s appendix model

Goodman’s first suggested attack asks whether L2 holds in Bacon’s appendix model when the pure unary operators are exactly the denotations of closed terms containing only logical vocabulary. Isabelle answers this question negatively.

Define the unary operator

$$Z(P) = \{ w : P(w \frown 0) \text{ and } P(w \frown 1) \text{ have different truth values} \}.$$

There is a closed term containing only logical vocabulary whose denotation is exactly Z . Hence Z is one of the pure unary operators denoted by closed terms containing only logical

vocabulary. Isabelle proves:

- (i) Z is surjective;
- (ii) $Z(P) = Z(\neg P)$, so Z is not injective;
- (iii) Z is right-cancellative on the denotations of closed logical terms;
- (iv) $Z \notin G$.

Surjectivity is witnessed explicitly by

$$P_S = \{v \smallfrown 0 : v \in S\}, \quad Z(P_S) = S.$$

Right-cancellativity implies that Zp is fun' whenever p is fun' . Choose such a p . Then

$$\text{id}(Zp) = Zp = Z(p),$$

although $\text{id} \not\approx Z$, since Z is not reversible. This is an explicit counterexample to L2. Since strong L2 entails weak L2, it fails in this model as well.

3.3 What this advances

This settles the first open problem in Goodman's list: L2 is false in Bacon's appendix model under the proposed interpretation of purity. The earlier Isabelle analysis had shown that no counterexample occurs among identity, necessity, possibility, constant truth, and constant falsity; Z is the first verified nonreversible closed logical operator that nevertheless preserves the distinctions relevant to fun' .

The result does not prove the consistency of PP. Bacon's appendix model still fails PP at the unary-operator type. What it shows is that failure of L2—one of the two escape routes forced by T6—is mathematically available in Bacon's own semantic setting. It also shows that a negative solution cannot treat L2 as automatic merely because purity is interpreted by closed logical definability.

4 Audit of the object-language theorems

Item	Status	Result
T1	Verified	Under zeroary Exhaustion, every pure proposition is truth or falsity. Consequently the biconditional operators are identity and negation, and WI collapses to Inv.
T2a	Verified	fun' is closed under every pure reversible operator, hence under negation.
T2b	Verified	Truth and falsity are not fun' . A fun' witness is distinct from truth, falsity, and its negation.
T2c	Verified, stronger	Every pure proposition is possibly identical to a fun' witness. The proof does not use PP.
T2d	Verified, stronger	Possible purity of the witness follows without Persistence, although Goodman's displayed route uses it.
T2e	Verified	The proposition that the witness is noncontingent is false but possible.
T2f	Verified conditionally	The six displayed propositions are pairwise distinct under the explicit $\text{fun}'(r)$ antecedent. The antecedent is essential.
T3	Repaired	The advertised proof has a modal gap: it reaches possible identity where identity is required. The theorem follows after adding zeroary Exhaustion, or from a weaker pure-identity rigidity principle. A two-world modal abstraction verifies that the advertised intermediate principles do not by themselves close the gap.
T4	Verified, stronger	For each closed pure $C : t \rightarrow (t \rightarrow t)$, $\sim \text{fun}'_{t \rightarrow t}(C(r))$. The proof uses only the purity schema and application closure; neither PP nor $\text{fun}'(r)$ is needed.
T5	Verified conditionally	The assumptions displayed in the notes, together with $\text{fun}'(r)$, yield a fun' proposition distinct from r and $\sim r$. This does not establish consistency of the antecedent.

Item	Status	Result
T6	Verified	All four contradiction routes are formalized: L2+Inv, L2+Wl, L2+TU, and strong-L2+RS. Recombination, the proposition-type instance of Exhaustion, and unique fundamentality derive, rather than assume, the required fun' witness where appropriate. The advertised T6-Wl master claim is also derived directly from its stated stock, independently of explosion.
T7a	Verified	For the liar family, the identity-indexed member is false, while some member indexed by a pure reversible operator is true.
T7b	Underspecified	The notes do not give the kind-level diagonal term, its binders, or precise definitions of “fixed” and “shifted” point. There is no unique theorem to encode.
T8	Verified	The five base kinds are pairwise distinct; L2 gives kind uniqueness; and the assumptions displayed in the notes prove 31 pairwise-distinct pure unary operators and 31 pairwise-distinct propositions.
T9	Verified at the counting level	The PC specification makes the map from sets of kinds to pure operators injective. Representation of the members of each kind by members of G gives an injective code of that kind into G . These results imply $\mathcal{P}(K) \leq K \times G .$ Classical cardinal arithmetic then gives $K \text{ finite} \quad \vee \quad \mathcal{P}(K) \leq G .$ Instantiating the abstract types uniformly across Goodman's full typed object language remains separate.

5 Audit of the model-theoretic claims

Item	Status	Result
M1	Exact identification; exclusion conditional	At proposition type, the pure propositions in Bacon's model are exactly the two invariant propositions, and the operator testing for membership in that class is the noncontingency operator. The exact footnote-59 liar is typed, beta-reduced, and proved pure from PP plus Purity of Fun; given QSS, this yields a contradiction. At the next type, the infinitary join exists in the full function domain and has exactly the pure unary operators as its extension. It interprets Pure at that type. PP holds there exactly when this operator is itself in the next pure stock. The current formalization does not prove directly that it is absent from the next pure stock in Bacon's appendix construction. It proves that absence only from the additional PP-diagonal and QSS assumptions. Completeness of the function domain therefore supplies the join, but does not by itself settle its purity.
M2	Verified	Goodman's construction $T \mapsto G_T$ is a bijection between arbitrary sets of propositions and invariant unary operators. Cantor's theorem yields more invariant operators than propositions, so evaluation at one fundamental proposition cannot be injective on all invariants. The invariance reading of purity therefore violates QSS.
M3	Verified relative to the pure operators	Relative to the chosen stock of pure operators, fun' is equivalent to freeness against nonzero pure laws. Such propositions have both extreme views. A countable Boolean stock has a free generator, while the resulting fun' class is topologically meager in the product topology.

Item	Status	Result
M4	Conditional model-theoretic theorem	Preimage heredity is proved. The lifted-branch witness is fun' and outside the reversible orbit under explicit assumptions: necessitated QSS, function-space membership and invariance of every operator in the pure stock, and presence of identity, zero, and one. The theorem has not yet been instantiated for Bacon's stock of operators denoted by closed terms containing only logical vocabulary. The multiple-fundamental "wide Fun" argument remains prose-level.

Item	Status	Result
M5	Verified and repaired; one generalization open	The operator $f = G_T$ obtained by exchanging s and s' is an invariant involution violating TU, WI, and Inv. The notes' subsequent countability argument is false: both the orbit and the displayed world-indexed family are countable. We replace it with a uniform diagonal construction. For every candidate R, it produces a two-element pair outside every view of R, whose proper views are all empty or universal. The induced invariant nonidentity transposition fixes R. If it and identity are both included among the pure operators, R ceases to be fun', proving the corrected pre-rebuild QSS obstruction. The rebuilding question is now settled. Isabelle constructs the least application-closed pure stock containing every closed logical denotation and the displayed f, proves that f is a pure self-inverse operator in the modalized function domain, and rebuilds a fundamental proposition that supplies Recombination and fun'-separation at every world. The rebuilt interpretation validates Bacon's Recombination background. It does not validate PP; that would require the classifier of the enlarged pure stock itself to belong to the next pure stock. The displayed collision calculation is also verified: the closed operator $\lambda p (p \leftrightarrow \text{NC}(p))$ takes the same value at $\text{NC}(r)$ and truth, although $\text{NC}(r) \neq \text{truth}$, whenever $\text{fun}'(r)$. Extending this argument uniformly to merely existentially given invertibles remains open.

Item	Status	Result
M6	Conditional model-theoretic theorem	Under the same explicit assumptions—necessitated QSS, membership of every pure operator in Bacon’s function domain, and invariance—distinct substitutions are separated by a fun' proposition. With identity, zero, and one, a strict-inclusion fun' pair blocks the indicated joint assignment. Single-coordinate arbitrary realization fails independently by countability. These results have not all been instantiated for Bacon’s stock of operators denoted by closed terms containing only logical vocabulary.
M7	Verified, with an open instance	The Tarski diagonal lies outside the range of the enumerated closed terms. For arbitrary invariant operators of Goodman’s form G_T , every proposition is reachable from r if and only if the orbit map $i \mapsto i \cdot r$ is injective. Whether Bacon’s glued r has this property is construction-sensitive and open.
10.1	Verified, qualified	Bacon’s rebuilt-family theorem is proved in a set-theoretic formalization of his appendix model. At type Ind , equality is literal identity, and the theorem’s hypotheses therefore force the family to be constant; the theorem cannot supply a family of distinct individual denotations.

6 Corrections and clarifications to the notes

6.1 The modal gap in T3

The advertised argument uses necessitated QSS and Persistence to move from a possible fundamental role to an injectivity claim. The formal derivation reaches a statement of the form

$$\Diamond(Y = Z),$$

where the desired conclusion requires $Y = Z$. No rule of H licenses that transition. The gap can be repaired in either of two principled ways.

1. Zeroary Exhaustion collapses pure propositions to truth and falsity, which supplies the required rigidity.
2. One may assume only the weaker principle that possible identity between pure operators implies identity.

The unrestricted version of the second principle is false. Thus the repair must remain restricted to the pure case.

6.2 The countability error in M5

The notes propose choosing a pair $\{s, s'\}$ outside the orbit of r because the orbit is countable and, it is said, there are uncountably many world-indexed pairs. But Bacon's worlds are finite sequences of natural numbers, hence countable. The displayed family of pairs is therefore countable as well. Countability alone proves nothing.

The replacement is direct diagonalization. Enumerate the views of R by the worlds. Define a proposition s_R whose truth at the singleton $[n+2]$ disagrees with the corresponding singleton's membership in the n -th view of R , and let

$$s'_R = s_R \cup \{\langle \rangle\}.$$

Membership above a nonactual world depends only on the final symbol of its sequence. Hence every nonactual view of s_R and s'_R is empty or universal, while both members differ from every view of R . Exchanging their membership in the set T_0 of true propositions yields an invariant nonidentity operator $f_R = G_T$ with

$$f_R(R) = R.$$

If f_R and identity are both included among the pure operators, evaluation at R is not injective. This is exactly the pre-rebuild QSS obstruction the M5 strategy needs.

The separate rebuilding step is also now verified. Instead of retaining this old R , the rebuilt model chooses a new fundamental proposition generic for the enlarged application-closed stock containing Goodman's displayed f . That proposition again satisfies Recombination and fun' -separation.

6.3 The scope of several semantic claims

The M4 and M6 theorems are genuine mathematical results, but at present they are conditional. They assume that every operator counted as pure belongs to Bacon's function domain and is invariant, together with necessitated QSS and, for some results, the presence of identity and the two constant operators. The proofs do not yet instantiate these assumptions for Bacon's complete stock of operators denoted by closed terms containing only logical vocabulary. We therefore do not count them as unconditional verifications of the corresponding claims about Bacon's particular glued model.

7 New theorems advancing the consistency question

The audit was not merely defensive. The formalization has produced several theorems not stated in the original notes. These results materially clarify the logical relations among Goodman's central claims.

7.1 Recombination, Exhaustion, and the existence of a fun' proposition

Theorem 1. *Unary Recombination yields the pointwise QSS core. Adding zeroary Exhaustion yields the necessitated universal form needed for QSS. Together with unique proposition-level fundamentality, the resulting stock proves that a fun' proposition exists.*

This theorem removes the previously assumed existential fun' premise from the premises used in T6. It also makes explicit exactly where Exhaustion, rather than Recombination, enters the object-language argument.

7.2 An exact semantic consequence of T6

Theorem 2. *Any model validating Bacon's proof theory, Recombination, the proposition-type instance of Exhaustion, unique proposition-level fundamentality, and PP must falsify global L2 or global TU:*

$$\sim \text{GValid}(\text{L2}) \quad \vee \quad \sim \text{GValid}(\text{TU}).$$

Moreover, some world in the constructed model falsifies one of the two closed formulas.

Here $\text{GValid}(A)$ means that the closed formula A is true at every world of the model. This formalizes the central dilemma in Goodman's notes. It does not select a disjunct. A positive model may exploit either cross-kind collisions on fun' inputs or a pure reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping.

7.3 The rebuilt M5 model

Theorem 3 (Completion of the rebuilding step). *There is a typed interpretation of Bacon's Recombination background whose pure denotations form the least application-closed stock containing all closed logical denotations and Goodman's displayed exotic operator f . That stock is countable. The operator f is in the modalized unary function domain, commutes with taking views, is self-inverse, is not truth-uniform, and is not a biconditional operator. A rebuilt fundamental proposition supplies Recombination and fun' -separation at every world.*

This discharges the rebuilding step that Goodman's notes left “modulo Theorem 10.1.” It verifies the intended M5 independence result for Bacon's Recombination background. It does not provide a model of PP: PP would add the further requirement that the classifier of this enlarged pure stock itself occur at the next type.

7.4 A uniform pre-rebuild M5 obstruction

Theorem 4 (Uniform version of the M5 construction). *For every candidate proposition R , there is a pair $\{s_R, s'_R\}$ disjoint from the set of all views of R . Every nonactual view of either member is empty or universal. The associated invariant transposition fixes R and is not identity.*

This is stronger and cleaner than the original M5 selection argument. It shows that an exotic invariant operator fixing any preassigned R can be constructed without cardinal assumptions. It shows that the pre-rebuild failure is caused by keeping the old fundamental proposition fixed while enlarging the pure operators. The preceding theorem shows how rebuilding the fundamental proposition avoids that failure.

7.5 The T9 and M7 reductions

Theorem 5 (T9 dichotomy). *If PC supplies an injection $\mathcal{P}(K) \rightarrow P$ into the pure unary operators, and every $\approx$ -class injects into G , then*

$$K \text{ is finite} \quad \vee \quad |G| \geq 2^{|K|}.$$

Theorem 6 (Invariant reachability). *Every proposition is reachable from r by an invariant operator if and only if*

$$i \mapsto i \cdot r$$

is injective.

The second theorem turns the open invariant form of Fundamental Completeness into a precise property of the chosen fundamental proposition. It also confirms that the issue depends on Bacon's particular construction rather than following from general facts about substitutions.

8 Where the open question now stands

Goodman's question asks whether Bacon's background theory, unique proposition-level fundamentality, and PP have a model. The verified results clarify both possible answers without deciding between them.

Affirmative route. A positive answer requires a model of the full theory plus PP, with purity interpreted at every relevant type. T6 shows that such a model must falsify L2 or TU. Thus it must contain either pure operators from distinct $\approx$ -classes that agree on suitable fun' propositions, or a pure reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping. Bacon's substitution-structure models do not already supply the desired example: M1 locates their first obstruction to PP, while M2 shows that identifying purity with invariance is incompatible with QSS. The operator Z shows that the operators denoted by Bacon's closed logical terms already take the $\neg\text{L2}$ branch, but the model's independent failure of PP prevents this from being a positive solution.

Negative route. The most direct negative route is the one already isolated in Goodman's notes: derive enough of L2 and one of Inv, WI, TU, or RS from the remaining principles, and apply T6. The formalization verifies the relevant T6 implications and removes the need to assume a fun' witness separately. The Z counterexample shows that Bacon's base theory and the interpretation of purity by denotations of closed logical terms do not by themselves secure L2. Any negative proof along this route must therefore use PP or some further principle not validated by that model.

Relative plausibility. The formal work does not establish that either answer is more likely. It shows exactly what each answer must overcome. A positive model must depart from Bacon's familiar substitution structures in one of the ways forced by the T6 disjunction. A negative proof must promote one of Goodman's model-sensitive principles, especially L2 or TU, to a consequence of the background theory itself.

9 Remaining verification work on the notes

The following items should not be reported as verified:

1. T7b, until a precise kind-level diagonal formula is supplied.
2. The multiple-fundamental “wide Fun” discussion in M4, which does not state a precise selection condition on Fun or a precise Separated Structure formula.
3. The proposed extension of the verified M5 collision calculation from its displayed operator to merely existentially given invertibles.

10 Recommended next steps

The following tasks arise directly from the unresolved or qualified claims in Goodman’s notes.

1. **Obtain determinate formulations.** Obtain an exact T7b statement, an exact formulation of the wide-Fun discussion before encoding them.
2. **Generalize the M5 collision method.** Determine whether the displayed noninjectivity calculation can be extended to invertible operators given only existentially, as would be required for the proposed PP-derived use of the method.
3. **Test Goodman’s negative route.** In light of the Z counterexample, determine whether L2 follows only after adding PP or some independently acceptable principle not valid in Bacon’s appendix model. Together with TU, WI, Inv, or the strong-L2+RS route, the verified T6 theorems then give the corresponding inconsistency result.
4. **Keep model claims and proof-theoretic claims separate.** In particular, do not promote the conditional M4/M6 stock theorems to claims about Bacon’s appendix model until their stock premises are instantiated.

11 Reproducibility

The repository’s three-layer organization and current build commands are documented in `README.md` and `docs/REPOSITORY_STRUCTURE.md`. We do not reproduce that implementation documentation here.

The controlling verification ledger is

`reports/GOODMAN_COMPLETE_VERIFICATION_MATRIX_2026-07-27.md`.

The dedicated audit theory is

`reports/audit_goodman_complete/Audit_Goodman_Complete.thy`.

The verification matrix gives, for each claim, the exact Isabelle theorem and the qualification under which it has been proved. The project status and handoff files identify the source theory corresponding to each result.

The appropriate overall verdict is therefore: *the formalization verifies, corrects, or precisely qualifies every determinate claim in Goodman's notes, including two substantive repairs. It also completes the rebuilding step in M5, refutes L2 in Bacon's appendix model under the proposed interpretation of purity, and proves the exact T6 semantic consequence and several strengthening results directly relevant to the notes. The remaining underspecified and model-dependent claims are explicitly identified. Goodman's consistency question itself remains open.*